

CALIBRATION KIT

Model ST-85054D18 DC-18GHz Type N SOLT Calibration Kit



INSTRUMENT LIST

Item No.	Instrument	Model Number	Quantity
1	Open (Male)	ST-85054D18-KM	1
2	Open (Female)	ST-85054D18-Kf	1
3	Short (Male)	ST-85054D18-DM	1
4	Short (Female)	ST-85054D18-DF	1
5	Load (Male)	ST-85054D18-LM	1
6	Load (Female)	ST-85054D18-LF	1
7	Storage Box 275*200*810mm		1
Optional accessories			
1	Adaptor: N(M)-N(M)	Optional	
2	Adaptor: N(F)-N(F)	Optional	
3	Adaptor: N(M)-N(F)	Optional	
4	Torque wrench: 19mm(3/4in), 1.35Nm	Optional	

SPECIFICATIONS

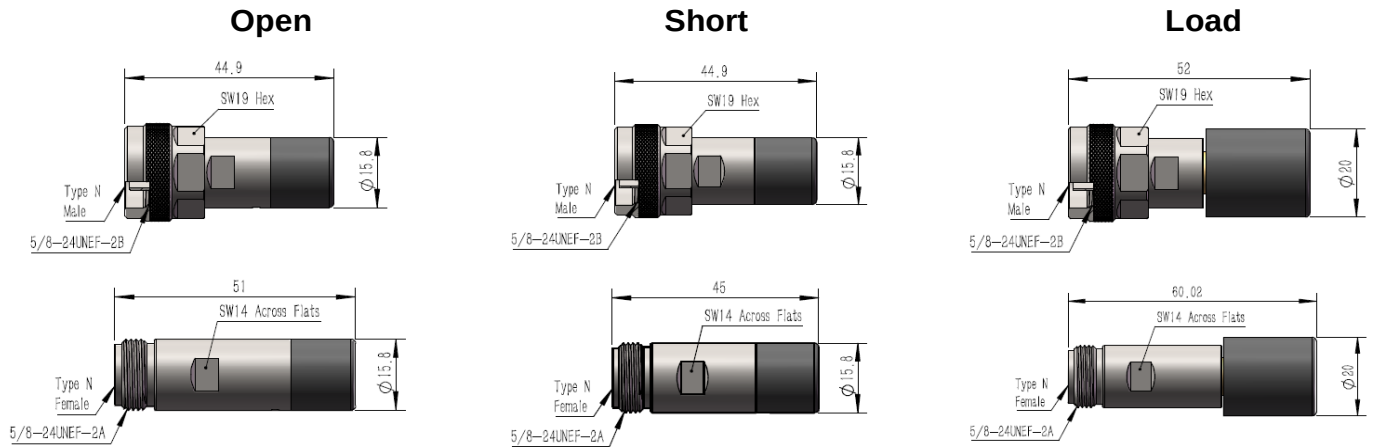
Instrument	Model Number	Frequency Range	Specifications
Open (Male)	ST-85054D18-KM	DC-18GHz	Phase deviation: $\leq \pm 1.0^\circ$ *
Open (Female)	ST-85054D18-Kf		Phase deviation: $\leq \pm 1.0^\circ$ *
Short (Male)	ST-85054D18-DM		Phase deviation: $\leq \pm 1.0^\circ$ *
Short (Female)	ST-85054D18-DF		Phase deviation: $\leq \pm 1.0^\circ$ *
Load (Male)	ST-85054D18-LM		VSWR: ≤ 1.048
Load (Female)	ST-85054D18-LF		VSWR: ≤ 1.048

CALIBRATION KIT

SURFACE TREATMENT

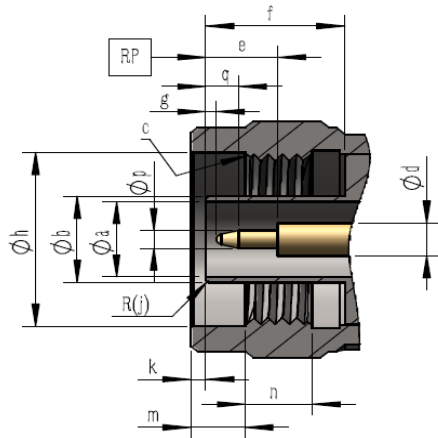
Instrument	Inner Conductor	Outer Conductor
Open	Beryllium bronze, gold-plated	SUS303 stainless steel passivated
Short	Beryllium bronze, gold-plated	SUS303 stainless steel passivated
Load	Beryllium bronze, gold-plated	SUS303 stainless steel passivated

OUTDRAWING:



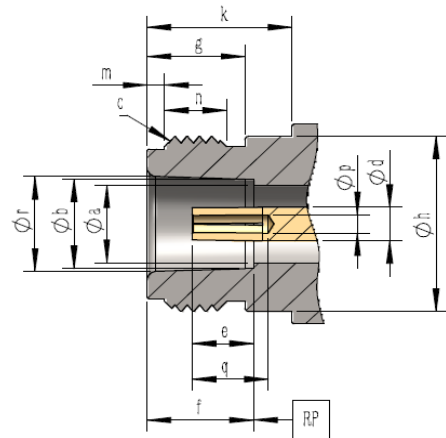
INTERFACE STANDARD:

Type N Male



	mm		in	
	Min.	Max.	Min.	Max.
a	7.00 Ref.		0.2756 Ref.	
b	7.98	8.02	0.314	0.316
c			5/8-24UNEF-2B	
d	3.04 Ref.		0.120 Ref.	
e	5.26	5.36	0.207	0.211
f	9.25	---	0.364	---
g	0	1.57	0	0.062
h	16.0	---	0.630	---
j	0.15	---	0.006	---
k	0.41	1.52	0.016	0.060
m	4.013	4.267	0.158	0.168
n	4.5	---	0.177	---
p	1.64	1.66	0.0646	0.0654
q	2.79	3.56	0.110	0.140

Type N Female



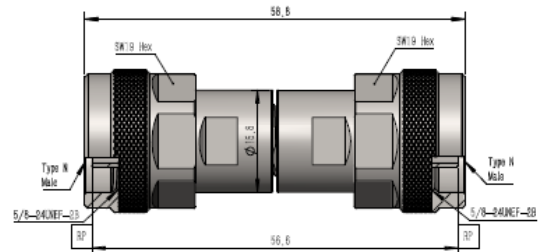
	mm		in	
	Min.	Max.	Min.	Max.
a	6.98	7.02	0.275	0.276
b	8.027	8.13	0.316	0.320
c			5/8-24UNEF-2A	
d	3.04 Ref.		0.120 Ref.	
e	5.16	5.26	0.203	0.207
f	9.05	9.15	0.356	0.360
g	6.76	---	0.266	---
h	---	15.93	---	0.627
k	10.72	---	0.422	---
m	1.19	1.96	0.047	0.077
n	4.37	---	0.172	---
p	---	1.75	---	0.069
q	5.33	---	0.210	---
r	8.53	8.74	0.336	0.344

CALIBRATION KIT

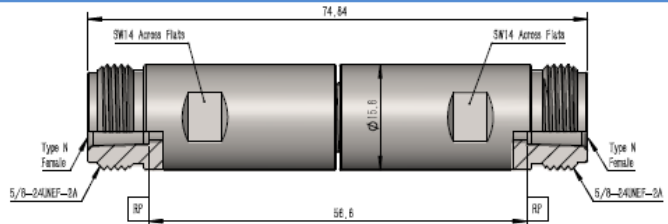
ADAPTOR:

Instrument	Frequency Range	Impedance	VSWR	Cycle Life	Temp Range
N(M)-N(M)	DC-18GHz	50Ω	≤1.06	> 2000 times	-55~+85℃
N(F)-N(F)			≤1.06		
N(M)-N(F)			≤1.06		
Material and Surface Treatment					
Outer Conductor		Inner Conductor		Insulator	Threaded sleeve
Stainless steel SUS303 passivated		Beryllium bronze gold-plated		PEEK	Stainless steel SUS303 passivated

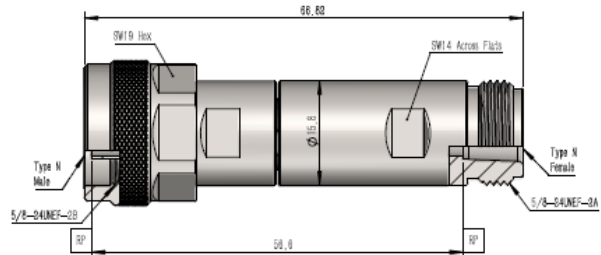
Type N (m) to Type N (m)



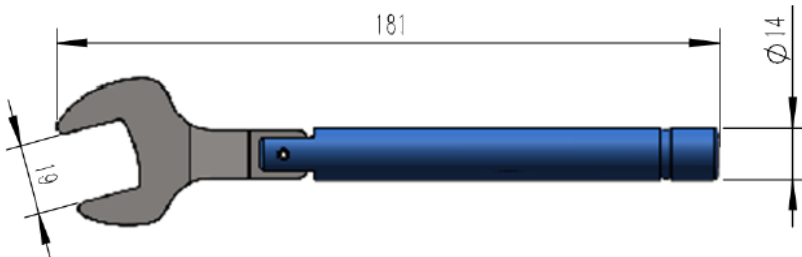
Type N (f) to Type N (f)



Type N (m) to Type N (f)



Torque wrench:



Opening size: 19mm (3/4 inch)

Torque setting: 1.35Nm (12 inch-lbs)

Wrench handle: Aluminum, anodized

Wrench head: Tool steel, nickel plated

CALIBRATION KIT

CALIBRATION PARAMETERS:

OPEN		Male	Female
Z0		50Ω	50Ω
Fringing Capacitances	89.939	104.13	104.13
	2536.7999	-1943.4008	-1943.4008
	-264.99	144.62	144.62
	13.4	2.2258	2.2258
Offset Delay (ps)		57.993	22.905
Offset length (mm)		17.386	6.867
Offset Loss	0.93	0.93	0.93
	0.0094	0.0037	0.0037
	2.7e-4	2.7e-4	2.7e-4

SHORT		Male	Female
Z0		50Ω	50Ω
Short Inductance	0.7563	-0.1315	-0.1315
	459.8799	606.2089	606.2089
	-52.429	-68.405	-68.405
	1.5846	2.0206	2.0206
Offset Delay (ps)		63.078	27.99
Offset length (mm)		18.91	8.391
Offset Loss	1.1273	1.13651	1.13651
	0.0123	0.0055	0.0055
	3.3e-4	3.3e-4	3.3e-4

LOAD	
Z0	50Ω
Offset Delay	0ps
Offset length	0mm
Offset Loss	0ohm/Sec 0dB/(GHz ^{1/2}) 0dB/mm

LOAD	
Z0	50Ω
Connector Port1	Type N female
Connector Port2	Type N male
Offset Delay	0ps
Offset length	0mm
Offset Loss	0ohm/Sec 0dB/(GHz ^{1/2}) 0dB/mm